

Supplementary Material

Emergent Fast-Slow Dynamics in Multi-Agent Q-Learning for Networked Stochastic Games

Yuxin Geng, Wolfram Barfuss, and Xingru Chen

1 Theoretical Derivation

In this section, we present the full derivation of the stochastic differential equation (SDE) and Fokker-Planck equation (FPE) for the dynamics of Q-values and states in stochastic games.

1.1 SDE

For a representative agent with Q-value vector $\mathbf{Q}$, the temporal evolution of the Q-value vector is given by the following SDE:

$$d\mathbf{Q} = \boldsymbol{\mu}(\mathbf{Q})dt + \sqrt{\boldsymbol{\Sigma}(\mathbf{Q})}d\mathbf{W}, \quad (1)$$

where $\boldsymbol{\mu}(\mathbf{Q})$ is the drift term and $\boldsymbol{\Sigma}(\mathbf{Q})$ is the diffusion term.

The drift term $\boldsymbol{\mu}(\mathbf{Q})$ is computed as the average change of Q-values per unit time, which is computed by averaging over the focal agent's probability of selecting action a in state s , the neighbor distribution $p_t(\tilde{\mathbf{Q}}|\mathbf{Q}, s)$, the neighbor's action $\tilde{a}$, and the transition kernel T :

$$\begin{aligned} \mu_t(\mathbf{Q}, s, a) = & \underbrace{\alpha}_{\text{learning rate}} \cdot \underbrace{\left\{1 - [1 - p_t(s|\mathbf{Q})]^k\right\}}_{\text{prob of at least one interaction in } s} \cdot \underbrace{X(\mathbf{Q}, s, a)}_{\text{prob of selecting } a \text{ in } s} \\ & \cdot \int \cdots \int \sum_{\tilde{a}} \sum_{s'} \left[R(s, a, \tilde{a}) + \gamma \max_{\tilde{a}'} Q_{t+1}(s', \tilde{a}') - Q_t(s, a) \right] \\ & \cdot \underbrace{X(\tilde{\mathbf{Q}}, s, \tilde{a})}_{\text{neighbor's prob of selecting } \tilde{a} \text{ in } s} \cdot \underbrace{T(s, a, \tilde{a}, s')}_{\text{prob of transition from } s \text{ to } s' \text{ when selecting } a \text{ and } \tilde{a}} \cdot \underbrace{p_t(\tilde{\mathbf{Q}}|\mathbf{Q}, s) d\tilde{\mathbf{Q}}}_{\text{prob of having a neighbor with Q-value } \tilde{\mathbf{Q}} \text{ in } s}. \end{aligned} \quad (2)$$

Here, we denote the expected TD-error term on the right-hand side as $\bar{\delta}(\mathbf{Q}, s, a)$, so the above equation can be expressed as

$$\mu(\mathbf{Q}, s, a) = \alpha \cdot \left\{1 - [1 - p(s|\mathbf{Q})]^k\right\} \cdot X(\mathbf{Q}, s, a) \cdot \bar{\delta}(\mathbf{Q}, s, a). \quad (3)$$

The computation of the covariance matrix element $\Sigma[(s, a), (s', a')]$ can be divided into three cases:

For $s \neq s'$, since agents take actions independently, we have

$$\begin{aligned}
& \Sigma[\mathbf{Q}, (s, a), (s', a')] \\
&= \mathbb{E}[(Q_{t+1}(s, a) - Q_t(s, a))(Q_{t+1}(s', a') - Q_t(s', a'))] \\
&\quad - \mathbb{E}[Q_{t+1}(s, a) - Q_t(s, a)] \cdot \mathbb{E}[Q_{t+1}(s', a') - Q_t(s', a')] \\
&= X(\mathbf{Q}, s, a) \left\{ 1 - [1 - p(s|\mathbf{Q})]^k \right\} X(\mathbf{Q}, s', a') \left\{ 1 - [1 - p(s'|\mathbf{Q})]^k \right\} \cdot [\alpha \bar{\delta}(\mathbf{Q}, s, a) \alpha \bar{\delta}(\mathbf{Q}', s', a')] \\
&\quad - \mu(\mathbf{Q}, s, a) \mu(\mathbf{Q}', s', a') = 0.
\end{aligned} \tag{4}$$

For $s = s'$, and $a \neq a'$, we have $\mathbb{E}[(Q_{t+1}(s, a) - Q_t(s, a))(Q_{t+1}(s, a') - Q_t(s, a'))] = 0$, since at most one of $Q_{t+1}(s, a)$ and $Q_{t+1}(s, a')$ is updated in one time step:

$$\begin{aligned}
\Sigma[\mathbf{Q}, (s, a), (s, a')] &= \mathbb{E}[(Q_{t+1}(s, a) - Q_t(s, a))(Q_{t+1}(s, a') - Q_t(s, a'))] \\
&\quad - \mathbb{E}[Q_{t+1}(s, a) - Q_t(s, a)] \cdot \mathbb{E}[Q_{t+1}(s, a') - Q_t(s, a')] \\
&= -\mu(\mathbf{Q}, s, a) \mu(\mathbf{Q}, s, a').
\end{aligned} \tag{5}$$

For $s = s'$, and $a = a'$, we have

$$\begin{aligned}
& \Sigma[\mathbf{Q}, (s, a), (s, a)] \\
&= \mathbb{E}[(Q_{t+1}(s, a) - Q_t(s, a))^2] - \mathbb{E}[Q_{t+1}(s, a) - Q_t(s, a)]^2 \\
&= \alpha^2 X(\mathbf{Q}, s, a) \left\{ 1 - [1 - p(s|\mathbf{Q})]^k \right\} \cdot \int \cdots \int \sum_{\tilde{a}} \sum_{s''} \left[R(s, a, \tilde{a}) + \gamma \max_{a''} Q_{t+1}(s'', a'') - Q_t(s, a) \right]^2 \\
&\quad \cdot X(\tilde{\mathbf{Q}}, s, \tilde{a}) \cdot T(s, a, \tilde{a}, s'') \cdot p_t(\tilde{\mathbf{Q}}|\mathbf{Q}, s) d\tilde{\mathbf{Q}} \\
&\quad - \mu(\mathbf{Q}, s, a)^2.
\end{aligned} \tag{6}$$

As a result, the elements of the drift vector $\boldsymbol{\mu}$ are of order $O(\alpha)$, and the elements of the covariance matrix Σ are of order $O(\alpha^2)$, which leads to the identification of different time scales between state mixing and Q-value updates, enabling their separation under $\alpha \ll 1$.

2 Additional Validation through Agent-Based Simulations

Complementary to the main text, we perform additional agent-based simulations to validate the theoretical results. Except for the parameters listed in the figure panel, we use the same parameters as in Figure 3 of the main text. In the figure, the markers represent theoretical predictions while the curves show the agent-based simulation results averaged over 10 runs. We omit variance indicators in the plot since the variance across runs is consistently small. As shown in the following figures, the robustness observed throughout the parameter sweep supports the practical relevance of our framework in diverse multi-agent learning settings.

2.1 Validation of inverse temperature β

In Fig. 1, we present the simulation results and theoretical predictions for different values of β . For $\beta = 1$, agents become cooperative in both states, as the probability of selecting action a_1 is

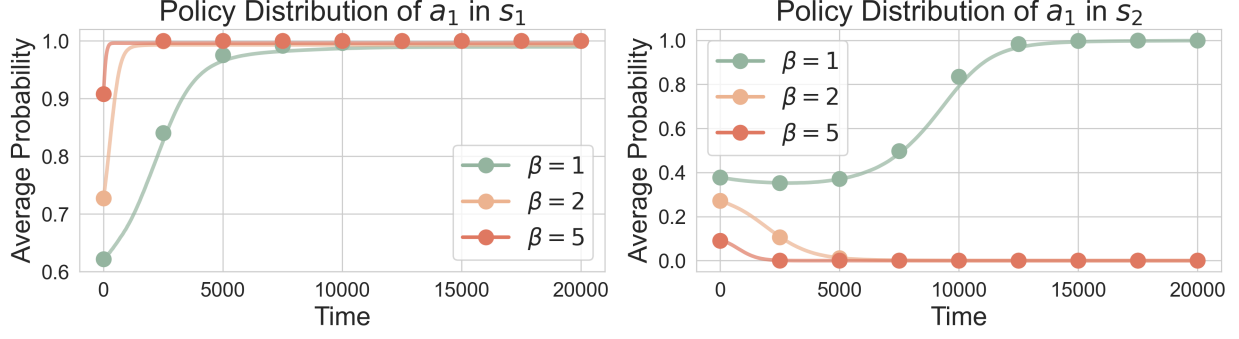


Figure 1: Simulation results and theoretical predictions for different values of β .

close to 1. Highly rational players with large β values fail to resolve the social dilemma, since once some agent defects and the system transitions to the degraded state s_2 , mutual defection tends to persist, causing the system to remain in s_2 .

In contrast, for $\beta = 2$ and $\beta = 5$, agents become cooperative in the prosperous state s_1 but become defective in the degraded state s_2 . In such cases, less rational agents with lower β explore more frequently. This exploratory behavior enables them to discover and maintain the cooperative equilibrium across both states. Even when occasional defections cause the system to fall into s_2 , agents still tend to cooperate, and the system is more likely to recover and return to the prosperous state s_1 . As a result, exploratory agents successfully resolve the social dilemma.

2.2 Validation of transition probabilities p_1 and p_2

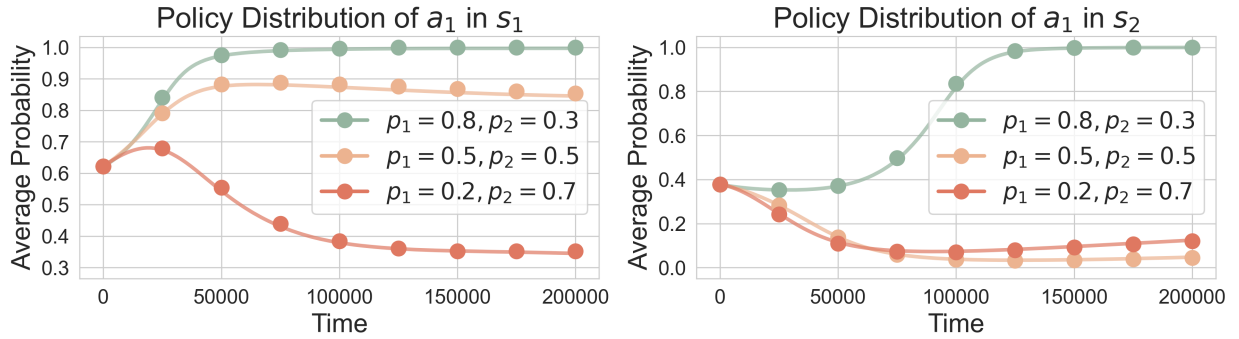


Figure 2: Simulation results and theoretical predictions for different transition probabilities p_1 and p_2 . We set $\alpha = 0.0001$.

In Fig. 2, we present the validation results for different transition probabilities p_1 and p_2 . With the increase of p_1 , the system is more likely to transition to the prosperous state s_1 when agents mutually cooperate. With the decrease of p_2 , the system is more likely to transition

to the degraded state s_2 when agents defect. As shown in the figure, agents become more cooperative when p_1 is large and p_2 is small. In such a scenario, defection drives the system into the degraded state s_2 , where the benefit b_2 is lower. Intuitively, although agents obtain higher immediate rewards by defecting, they lose more long-term benefits due to the transition to the degraded state s_2 . In other words, the state transition can be viewed as a punishment mechanism for defection, which promotes the emergence of cooperation by reducing future payoffs of short-sighted agents.